

PERCENTAGE & PROFIT-LOSS

Fully Solved Questions with Step-by-Step Explanation & Shortcut Tricks

How to use these notes: Every question is solved in full steps first, followed by a ✨ Trick box that gives the fastest exam method for that concept. Study the trick after understanding the full solution.

SECTION A — PERCENTAGE

Q1. If we express $12\frac{4}{15}\%$ as a fraction, then it is equal to:

(a) $46/375$ (b) $46/125$ (c) $23/250$ (d) None of these

$$12\frac{4}{15}\% = (12 \times 15 + 4)/15\% = 184/15\%$$

To convert % to a fraction, divide by 100: $(184/15) \div 100 = 184/1500$

Simplify by dividing numerator & denominator by 4: $184/1500 = 46/375$

Answer: (a) $46/375$

✨ **Trick / Shortcut**

To convert any mixed-number percentage $a(b/c)\%$ to a fraction: write it as an improper fraction over 100, i.e. $[(ac+b)/c] \div 100$.

Memorise common %-to-fraction conversions ($12.5\% = 1/8$, $16.66\% = 1/6$, $33.33\% = 1/3$, $6.25\% = 1/16$) — they make percentage calculations almost instant.

Q2. What is 10% of 20% of 25% of 100?

(a) 0.5 (b) 0.75 (c) 0.25 (d) 1.0

Convert every percentage to a fraction and multiply directly with 100:

$$10\% \text{ of } 20\% \text{ of } 25\% \text{ of } 100 = (10/100) \times (20/100) \times (25/100) \times 100$$

$$= 0.1 \times 0.2 \times 0.25 \times 100 = 0.005 \times 100 = 0.5$$

Answer: (a) 0.5

✨ **Trick / Shortcut**

'% of % of %' questions are just successive multiplication — convert each % to a decimal ($10\% \rightarrow 0.1$, $20\% \rightarrow 0.2$, $25\% \rightarrow 0.25$) and multiply straight through, no need to solve step by step separately.

Q3. If 25% of x = 30% of y, then find the value of x if y = 5000.

(a) 2000 (b) 3000 (c) 4000 (d) 6000

$$25\% \text{ of } x = 30\% \text{ of } y \rightarrow 0.25x = 0.30y$$

$$x = (0.30/0.25) \times y = 1.2y$$

$$x = 1.2 \times 5000 = 6000$$

Answer: (d) 6000

✨ **Trick / Shortcut**

Whenever ' $a\%$ of $x = b\%$ of y ', directly write $x = (b/a) \times y$ — no need to set up decimals each time, just take the ratio of the two percentages.

Q4. 20% of a number when subtracted from 108, gives the number itself. Find the number.

(a) 50 (b) 80 (c) 70 (d) 90

Let the number be N .

$$108 - 20\% \text{ of } N = N \rightarrow 108 - 0.2N = N$$

$$108 = N + 0.2N = 1.2N$$

$$N = 108 \div 1.2 = 90$$

Answer: (d) 90

✨ **Trick / Shortcut**

Convert the word problem into one clean linear equation in N first; keep the percentage as a decimal coefficient of N so it combines directly with the other N term.

Q5. Two students appeared at an examination. One of them secured 10 marks more than the other and his marks was 60% of the sum of their marks. The marks obtained by the better student are:

(a) 30 (b) 70 (c) 60 (d) 50

Let the sum of their marks = S .

Better student's marks = 60% of $S = 0.6S$; so the other student's marks = $0.4S$

Difference = $0.6S - 0.4S = 0.2S$, and this difference is given as 10

$$0.2S = 10 \rightarrow S = 50$$

$$\text{Better student's marks} = 0.6 \times 50 = 30$$

Answer: (a) 30

✨ **Trick / Shortcut**

When one part is ' $p\%$ of the total', the other part is automatically $(100-p)\%$ of the total — use this to write the difference directly as $(2p-100)\%$ of the total and solve for the total in one step.

Q6. A student has to obtain 33% of the total marks to pass. He got 125 marks and failed by 40 marks. The maximum marks are:

(a) 500 (b) 600 (c) 400 (d) 450

$$\text{Passing marks} = \text{marks obtained} + \text{marks short} = 125 + 40 = 165$$

Passing marks = 33% of Maximum marks

$$0.33 \times \text{Maximum} = 165 \rightarrow \text{Maximum} = 165 \div 0.33 = 500$$

Answer: (a) 500

✨ **Trick / Shortcut**

Standard formula: $\text{Maximum Marks} = (\text{Marks obtained} + \text{Marks short of passing}) \div (\text{Pass percentage})$. Apply directly without extra steps.

Q7. Mr. Ram is worried about the balance of his monthly budget. The price of petrol has increased by 50%. By what percent should he reduce the consumption of petrol so that he is able to balance his budget?

(a) 33.33 (b) 28.56 (c) 25 (d) 14.28

Expenditure = Price $\times$ Consumption must remain constant.

New price = old price $\times$ 1.5

For expenditure to stay the same: New consumption = Old consumption $\div$ 1.5 = Old consumption $\times$ (2/3)

Reduction in consumption = $1 - 2/3 = 1/3 = 33.33\%$

Answer: (a) 33.33

⚡ **Trick / Shortcut**

MASTER FORMULA for 'price up by x%, find reduction in consumption to keep expenditure same': $\text{Reduction}\% = [x / (100+x)] \times 100$.

Here: $[50 / (100+50)] \times 100 = 50/150 \times 100 = 33.33\%$. This formula also works in reverse for 'price reduced by x%, find increase in consumption to keep budget the same' — just use $\text{Increase}\% = [x / (100-x)] \times 100$.

Q8. A fruit seller had some apples. He sells 40% apples and still has 420 apples. What is the total number of apples he had originally?

(a) 800 (b) 700 (c) 500 (d) 600

If 40% apples are sold, 60% apples remain.

60% of Total = 420

Total = $420 \div 0.6 = 700$

Answer: (b) 700

⚡ **Trick / Shortcut**

Whenever 'x% is sold/used and the remainder is given', remember $\text{remainder}\% = (100 - x)\%$, and $\text{Total} = \text{Remainder} \div (\text{Remainder}\% \text{ as decimal})$.

Q9. A person multiplied a number by $3/5$ instead of $5/3$. What is the percentage error in the calculation?

(a) 52% (b) 64% (c) 56% (d) 52%

Correct value = $N \times 5/3$; Wrong value = $N \times 3/5$

Error = Correct - Wrong = $N(5/3 - 3/5) = N(25-9)/15 = 16N/15$

Percentage error = $\text{Error} \div \text{Correct value} \times 100 = (16N/15) \div (5N/3) \times 100$

= $(16/15) \times (3/5) \times 100 = (48/75) \times 100 = 64\%$

Answer: (b) 64%

⚡ **Trick / Shortcut**

Direct formula for 'multiplied by a/b instead of c/d ': $\% \text{ error} = |(c/d - a/b)| \div (c/d) \times 100$.

Even faster: since actual answer became $(a/b) \div (c/d)$ times the correct value, $\text{error}\% = [1 - (a/b) \div (c/d)] \times 100 = [1 - (3/5 \times 3/5)] \dots$ simplest is to just take the ratio of wrong-fraction to right-fraction: $(3/5) / (5/3) = 9/25 = 36\%$ of correct value obtained, so error = $100 - 36 = 64\%$.

Q10. Two friends, Akash & Beenu had some candies each. One of them had 15 candies more than the other. The candies with Akash was 60% of the total candies with them. How many candies did each have?

(a) 40, 25 (b) 47, 32 (c) 45, 30 (d) 49, 34

Let total candies = T. Akash = 60% of T = $0.6T$, Beenu = 40% of T = $0.4T$

Difference = $0.6T - 0.4T = 0.2T = 15$

T = 75

$$\text{Akash} = 0.6 \times 75 = 45; \text{Beenu} = 0.4 \times 75 = 30$$

Answer: (c) 45, 30

✨ **Trick / Shortcut**

Same pattern as Q5: one part is $p\%$ of total, so the difference between the two parts = $(2p-100)\%$ of total. Solve for total first, then find each part.

Q11. In an election between 2 candidates, Chaman gets 80% of the total valid votes. If the total votes were 12000, what is the number of valid votes that the other candidate Dhande gets, if 15% of the total votes were declared invalid?

(a) 1645 (b) 1545 (c) 1675 (d) 2040

$$\text{Valid votes} = (100 - 15)\% \text{ of } 12000 = 85\% \text{ of } 12000 = 10200$$

Chaman gets 80% of valid votes, so Dhande gets the remaining 20% of valid votes

$$\text{Dhande's votes} = 20\% \text{ of } 10200 = 2040$$

Answer: (d) 2040

✨ **Trick / Shortcut**

Two-stage percentage problems (Total $\rightarrow$ Valid $\rightarrow$ Candidate share): apply each percentage one after another on the correct base — never apply the candidate's % directly on the total votes; always go through the valid votes first.

Q12. Mr Amar spends 50% of his monthly income on household items and out of the remaining he spends 25% on travelling, 30% on entertainment, 15% on shopping and the remaining amount of Rs. 900 is saved. What is Mr Amar's monthly income?

(a) Rs. 6,000 (b) Rs. 12,000 (c) Rs. 9,000 (d) Cannot be determined

$$\text{Let monthly income} = I. \text{ After household expense: remaining} = 50\% \text{ of } I = 0.5I$$

Of this remaining $0.5I$, he spends $25\%+30\%+15\% = 70\%$, so savings = 30% of the remaining

$$\text{Savings} = 30\% \text{ of } 0.5I = 0.15I, \text{ and this equals Rs. 900}$$

$$0.15I = 900 \rightarrow I = 900 \div 0.15 = 6000$$

Answer: (a) Rs. 6,000

✨ **Trick / Shortcut**

In multi-stage spending problems, always multiply the percentages together to directly connect income to the final saved amount: here $0.5 \times 0.3 = 0.15$ of income = saving, skipping intermediate stages.

Q13. A landowner increased the length and the breadth of a rectangular plot by 20% and 30% respectively. Find the percentage change in the cost of the plot assuming land prices are uniform throughout his plot.

(a) 23% (b) 52% (c) 56% (d) None of these

Cost is proportional to Area = Length $\times$ Breadth

$$\text{New Length} = 1.20 \times \text{old Length}; \text{New Breadth} = 1.30 \times \text{old Breadth}$$

$$\text{New Area} = 1.20 \times 1.30 \times \text{old Area} = 1.56 \times \text{old Area}$$

$$\text{Percentage increase in cost} = (1.56 - 1) \times 100 = 56\%$$

Answer: (c) 56%

✨ **Trick / Shortcut**

SUCCESSIVE PERCENTAGE CHANGE FORMULA: if two quantities change by $x\%$ and $y\%$, the combined (product) change = $x + y + (xy/100) \%$.

Here: $20 + 30 + (20 \times 30)/100 = 20 + 30 + 6 = 56\%$. This formula is extremely fast for area/volume/price-quantity combined changes.

Q14. The population of a town is 6000. If the number of males increases by 10% and the number of females increases by 20%, then the population becomes 6800. Find the population of females in the town.

(a) 2500 (b) 3000 (c) 2000 (d) 3500

Let Males = M, Females = F; $M + F = 6000$

New population: $1.1M + 1.2F = 6800$

Substitute $M = 6000 - F$: $1.1(6000 - F) + 1.2F = 6800$

$6600 - 1.1F + 1.2F = 6800 \rightarrow 6600 + 0.1F = 6800 \rightarrow 0.1F = 200$

$F = 2000$

Answer: (c) 2000

✨ **Trick / Shortcut**

For 'two groups grow at different rates to a new total' problems, always form two equations: (i) old total, (ii) weighted new total — then eliminate one variable by substitution. Keep the smaller rate's coefficient with the substituted variable to simplify quickly.

Q15. The price of salt is reduced by 50% but, in spite of the decrease, Aayush ends up increasing his expenditure on salt by 50%. What is the percentage change in his monthly consumption of salt?

(a) +60% (b) -100% (c) +25% (d) 200%

Expenditure = Price $\times$ Consumption

New Price = $0.5 \times$ old Price; New Expenditure = $1.5 \times$ old Expenditure

New Consumption = New Expenditure $\div$ New Price = $(1.5 \times \text{old E}) \div (0.5 \times \text{old P}) = 3 \times (\text{old E}/\text{old P}) = 3 \times \text{old Consumption}$

Consumption becomes 3 times the original, i.e. it increases by $(3-1) \times 100\% = 200\%$

Answer: (d) 200%

✨ **Trick / Shortcut**

Whenever price and expenditure both change, use: New Consumption / Old Consumption = (New Expenditure / Old Expenditure) $\div$ (New Price / Old Price).

Here that ratio = $1.5 \div 0.5 = 3$, so consumption change = $(3-1) \times 100 = 200\%$ increase.

SECTION B — PROFIT & LOSS

Q16. A sold an article with 10% loss on the cost price. He bought the article at a discount of 20% on the labelled price. What would have been the percentage loss had he bought it at the labelled price?

(a) 34% (b) 18% (c) 16% (d) 28%

Let Labelled Price = L. A actually bought it at 20% discount, so his Cost Price = $0.8L$

He sold at 10% loss on his cost price: Selling Price = $0.9 \times 0.8L = 0.72L$

If instead he had bought it at the full labelled price, his Cost Price would be L, but the Selling Price is still the same, 0.72L

$$\text{Loss\% (on labelled price as CP)} = (L - 0.72L)/L \times 100 = 28\%$$

Answer: (d) 28%

✨ **Trick / Shortcut**

In these 'what-if he had paid the full price' problems, the Selling Price never changes — only the assumed Cost Price changes. Fix the SP first (from the real transaction), then compare it against the new hypothetical CP.

Q17. The owner of a cell phone shop charges his customer 28% more than the cost price. If a customer paid Rs 8,960 for the cell phone, then what was the cost price of the cell phone?

(a) Rs 7800 (b) Rs 7100 (c) Rs 6900 (d) Rs 7000

$$\text{Selling Price} = \text{Cost Price} + 28\% \text{ of Cost Price} = 1.28 \times \text{CP}$$

$$1.28 \times \text{CP} = 8960$$

$$\text{CP} = 8960 \div 1.28 = 7000$$

Answer: (d) Rs 7000

✨ **Trick / Shortcut**

Direct formula: $\text{CP} = \text{SP} \div (1 + \text{Profit\%/100})$. Just plug in numbers — no need to form an equation from scratch each time.

Q18. By selling an article for Rs. 480 a person lost 20%. For what price should he sell it to make a profit of 20%?

(a) Rs. 800 (b) Rs. 760 (c) Rs. 720 (d) Rs. 680

$$\text{SP} = 480 \text{ at a } 20\% \text{ loss, so } 480 = 80\% \text{ of CP}$$

$$\text{CP} = 480 \div 0.8 = 600$$

$$\text{Required SP for } 20\% \text{ profit} = 1.2 \times \text{CP} = 1.2 \times 600 = 720$$

Answer: (c) Rs. 720

✨ **Trick / Shortcut**

Quick two-step method for 'loss given, find SP for a profit': first $\text{CP} = (\text{given SP}) \div (1 - \text{Loss\%/100})$, then new $\text{SP} = \text{CP} \times (1 + \text{required Profit\%/100})$.

Shortcut ratio: $\text{New SP} = \text{Old SP} \times [(100 + \text{Profit\%}) / (100 - \text{Loss\%})] = 480 \times (120/80) = 720$ directly, without finding CP separately.

Q19. The profit earned after selling an article for Rs. 625 is the same as the loss incurred after selling the article for Rs. 435. The cost price of the article is:

(a) Rs. 520 (b) Rs. 530 (c) Rs. 540 (d) Rs. 550

$$\text{Let Cost Price} = C. \text{ Profit at Rs.625} = 625 - C. \text{ Loss at Rs.435} = C - 435$$

$$\text{Given: Profit} = \text{Loss, so } 625 - C = C - 435$$

$$625 + 435 = 2C \rightarrow 1060 = 2C \rightarrow C = 530$$

Answer: (b) Rs. 530

✨ **Trick / Shortcut**

Whenever 'profit at price P_1 equals loss at price P_2 ', the Cost Price is simply the average: $\text{CP} = (P_1 + P_2) \div 2$.

Here: $(625+435)/2 = 530$ directly, without forming the equation.

Q20. A merchant has 1000 kg of sugar, part of which he sells at 8% profit and the rest at 18% profit. He gains 14% on the whole. The quantity sold at 18% profit is:

(a) 500 kg (b) 600 kg (c) 400 kg (d) 640 kg

By alligation, ratio of quantity sold at 8% profit : quantity sold at 18% profit = $(18-14) : (14-8) = 4 : 6 = 2 : 3$

Total quantity = 1000 kg, divided in ratio 2:3 → total parts = 5

Quantity sold at 18% profit = $(3/5) \times 1000 = 600$ kg

Answer: (b) 600 kg

✨ **Trick / Shortcut**

This is an ALLIGATION problem on profit percentages (just like mixing two liquids): draw the alligation cross with 8% and 18% at the ends and 14% (overall) in the middle, then cross-subtract to get the ratio $4:6 = 2:3$ instantly.

Q21. A reduction of 20% in the price of rice enables a person to buy 2 kg more for Rs 30. Find the reduced and the original price per kg of rice.

(a) Rs 3 and $3\frac{3}{4}$ per kg (b) Rs 3 and $2\frac{3}{4}$ per kg (c) Rs 2 and 2 per kg (d) Rs 2 and $2\frac{3}{4}$ per kg

Let original price = P per kg. Reduced price = 0.8P per kg (20% reduction)

Quantity bought at reduced price – Quantity bought at original price = 2 kg (for the same Rs. 30)

$30/(0.8P) - 30/P = 2$

Multiply through: $37.5/P - 30/P = 2 \rightarrow 7.5/P = 2 \rightarrow P = 3.75$

Original price = Rs 3.75/kg = Rs $3\frac{3}{4}$ per kg; Reduced price = $0.8 \times 3.75 = \text{Rs } 3/\text{kg}$

Answer: (a) Reduced price = Rs 3/kg, Original price = Rs $3\frac{3}{4}$ /kg (i.e., Rs 3.75/kg)

✨ **Trick / Shortcut**

Formula for 'reduction of x% in price allows buying q kg more for Rs. R': Reduced Price = $(x/100) \times R \div q$.

Here: $(20/100) \times 30 \div 2 = 6 \div 2 = \text{Rs } 3/\text{kg}$ (reduced price) directly. Then Original = Reduced $\div 0.8 = 3.75$.

Q22. The price of tea is reduced by 40%. Now a person can buy 600 g more tea for Rs. 36. The original price of the tea per kilogram was:

(a) Rs. 20 (b) Rs. 30 (c) Rs. 40 (d) Rs. 42

Using the shortcut formula: Reduced price = $(\text{Reduction}\% \times \text{Amount spent}) \div (\text{extra quantity bought, in kg})$

Reduced price = $(40/100) \times 36 \div 0.6 = 14.4 \div 0.6 = \text{Rs } 24/\text{kg}$

Original price = Reduced price $\div (1 - 0.4) = 24 \div 0.6 = \text{Rs } 40/\text{kg}$

Answer: (c) Rs. 40

✨ **Trick / Shortcut**

Same formula as Q21: Reduced Price = $(x\% \times \text{Total Money}) \div (\text{Extra Quantity in kg})$, then Original Price = Reduced Price $\div (1 - x/100)$.

Always convert 'grams' to 'kg' first ($600 \text{ g} = 0.6 \text{ kg}$) before applying the per-kg formula.

Q23. Cost price of 100 books is equal to the selling price of 60 books. The gain percentage or loss percentage is:

(a) 60% (b) 67% (c) 65% (d) 66.66%

Let cost price of 1 book = Rs.1, so CP of 100 books = Rs.100

Given: CP of 100 books = SP of 60 books, so SP of 60 books = Rs.100, meaning SP of 1 book = $100/60$

Gain on 1 book = $SP - CP = 100/60 - 1 = 40/60 = 2/3$

Gain% = $(\text{Gain}/CP) \times 100 = (2/3) \times 100 = 66.66\%$

Answer: (d) 66.66%

✨ **Trick / Shortcut**

Direct formula: when CP of x articles = SP of y articles ($x > y$, a gain situation), $\text{Gain}\% = [(x-y)/y] \times 100$.

Here: $[(100-60)/60] \times 100 = (40/60) \times 100 = 66.66\%$. If $y > x$ instead, the same formula gives a Loss%.

Q24. An article is sold at a gain of 15%. Had it been sold for ₹27 more, the profit would have been 20%. The cost price of the article is:

(a) Rs 500 (b) Rs 560 (c) Rs 540 (d) Rs 520

Let Cost Price = C . SP at 15% gain = $1.15C$. SP at 20% gain = $1.20C$

Difference between the two SPs = 27 (given)

$1.20C - 1.15C = 27 \rightarrow 0.05C = 27$

$C = 27 \div 0.05 = 540$

Answer: (c) Rs 540

✨ **Trick / Shortcut**

Whenever two profit percentages on the same article differ by a known Rupee amount, that Rupee amount corresponds exactly to the DIFFERENCE of the two percentages: $\text{Rs. difference} = (\% \text{ difference} / 100) \times CP$.

Here: difference in % = $20 - 15 = 5\%$, so $CP = 27 \div (5/100) = 540$, found in one line.

Q25. Aalam sold two vehicles for ₹46000 each. If he gained 10% on the first and lost 10% on the other, then what is his percentage profit or loss in this transaction?

(a) 2% loss (b) 2% profit (c) 1% loss (d) 3% loss

When the SAME selling price gives $x\%$ profit on one item and $x\%$ loss on another, there is always an overall LOSS.

Overall Loss% = $(x/10)^2 = (10/10)^2 = 1^2 = 1\%$

Verify: $CP_1 = 46000/1.10 = 41818.18$; $CP_2 = 46000/0.90 = 51111.11$; Total CP = 92929.29; Total SP = 92000

Loss = $92929.29 - 92000 = 929.29$; Loss% = $(929.29/92929.29) \times 100 \approx 1\%$

Answer: (c) 1% loss

✨ **Trick / Shortcut**

GOLDEN FORMULA: If two articles are sold at the SAME price, one at $x\%$ profit and the other at $x\%$ loss, the overall result is always a LOSS of $(x/10)^2\%$.

This is a very common exam trap — students often (wrongly) assume profit and loss cancel out to 0%. They never do; there is always a net loss equal to $(x/10)^2\%$.